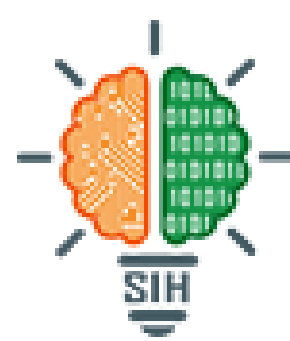




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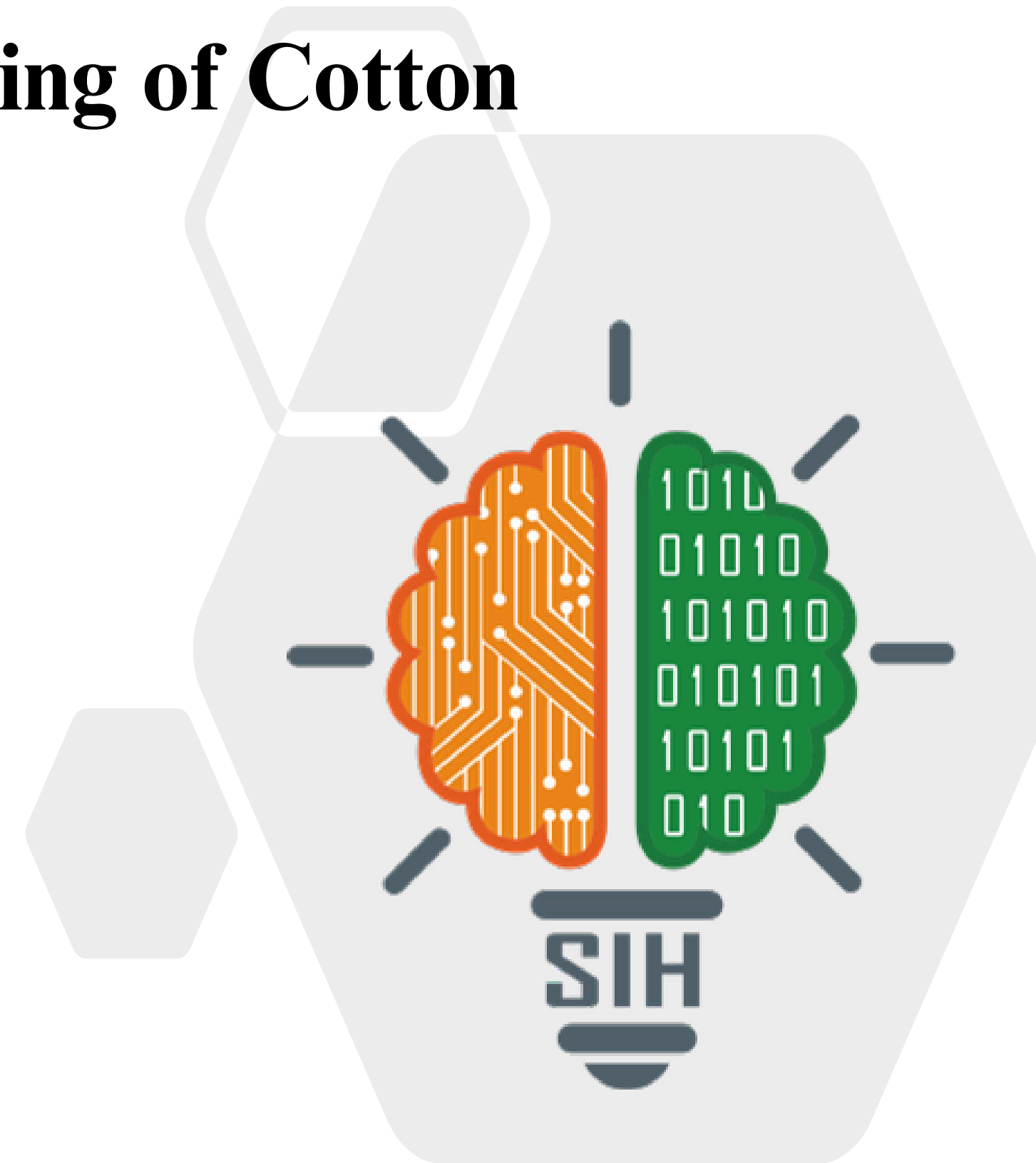
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HACKATHON
2025**

Low Cost Machinery for Picking of Cotton

- **Problem Statement ID – 25266**
- **Problem Statement Title -**

Low Cost Machinery for Picking of Cotton

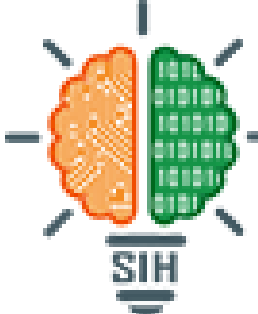
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Hardware**
- **Team Name (Registered on portal) Mindsmiths**



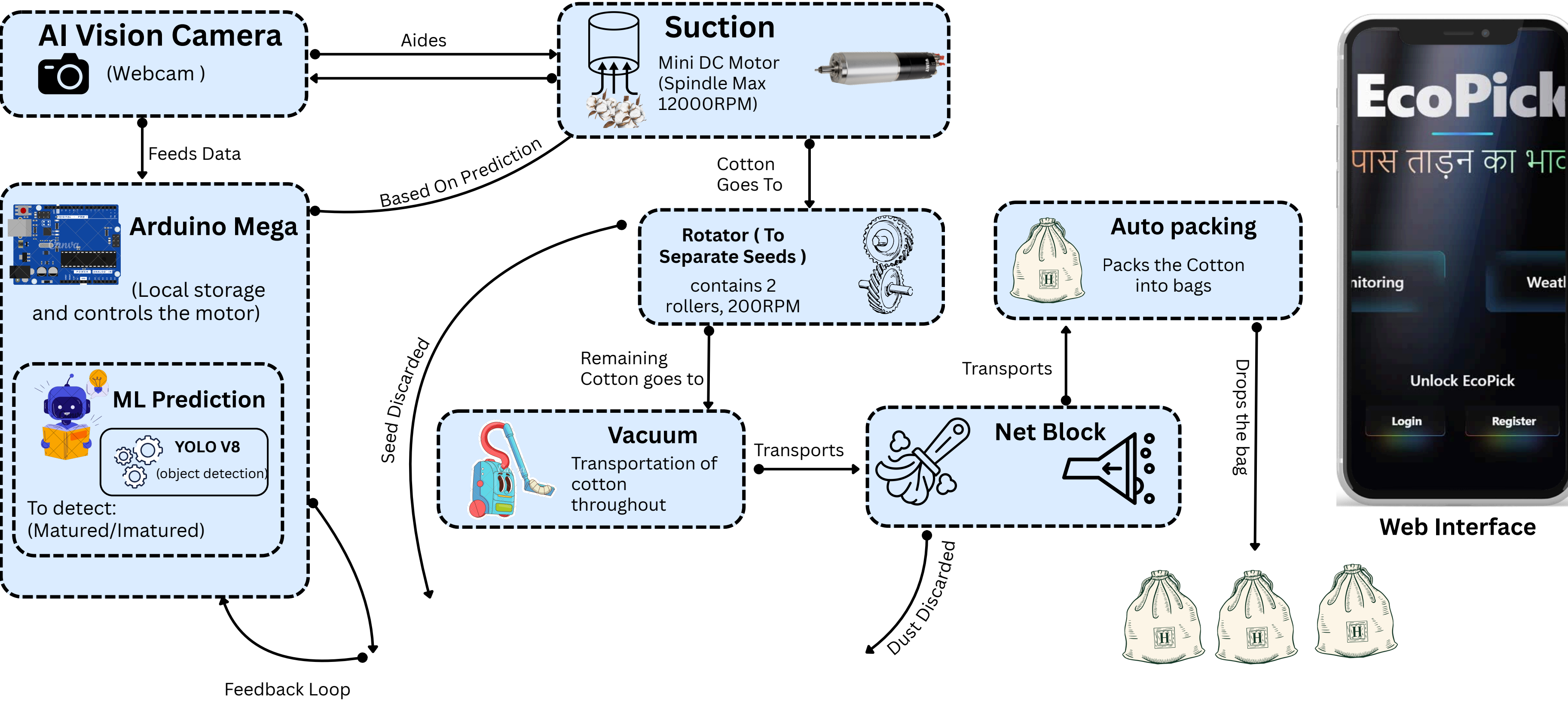


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TECHNICAL APPROACH



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FEASIBILITY AND VIABILITY

Technical Feasibility

- **Built using commercially available components** (Arduino Mega, Li-ion battery, solar panel, vacuum motor, 3D-printed frame).
- AI insights & IoT-enabled Wi-Fi connectivity for real-time pH, and yield data sync.
- **Low-power sensors** (NPK, moisture, pH) integrated without stressing battery life.

Economical Feasibility

Evaluation Parameter	Current Manual / Portable Picker	Proposed Smart System
Cost of Device	₹4,000 – ₹7,000	₹9,500 – ₹10,000
Manpower Required	2–3 workers	1 operator
Daily Picking Capacity	~17–25 kg	30–35 kg (selective, clean lint)
Trash Content	10–15 %	6–8 %
Labour Cost Reduction	—	≈ 60 %
Power Consumption	Battery only (limited)	Solar-battery hybrid (sustainable)

Maintenance & Safety Feasibility

- **Minimal maintenance** – few moving parts, easy to service with basic hand tools.
- **Electrostatic shielding**, dust filters, and insulated motor housing ensure user safety.
- **Predictive maintenance** via vibration and current sensing extends machine life.

Economic and Operational Viability

- **Total device cost under ₹20,000**, with daily operational cost < ₹60.
- **Payback within 1–2** harvests for 1–2 Hectare farms; ROI > 40%.
- **Lightweight backpack** design reduces operator fatigue by 30–40%.
- **Hip-mounted controller & multilingual mobile app.**

Implementation Viability

Factor	Support Element
Manufacturability	Simple low-cost machining or rural fab-lab assembly possible.
Component Availability	100 % components available in India.
Serviceability	Easy component replacement; no dependency on sophisticated tools.
Scalability	Modular add-ons (multi-nozzle picker, fiber sensor integration).
Deployment Readiness	Prototype ready for field testing; estimated commercial scaling possible within 12 months.

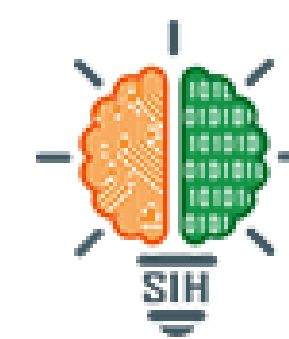
Environmental Viability

- **Low-heat, anti-static** materials meet agricultural safety standards.
- **Promotes climate-resilient** and low-carbon farming practices.
- **Supports Atmanirbhar Bharat & Digital India missions** through digital literacy and rural innovation.



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IMPACTS AND BENEFITS



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Cost Reduction

Reduces manual labor by 60%, cutting production costs significantly.



Affordability

Machine cost under Rs 20, 000 , making it accessible to small farmers.



Quality Improvement

AI-based selective picking improves lint purity and market premiums.

Economic Impact



AI Picking

AI accurately detects boll maturity for precise harvesting.



Yield Maps

Geo-tagged maps visualize productivity zones for better sowing.



Predictive Maintenance

Vibration/current sensing predicts maintenance needs, reducing downtime.



Scalable Design

Modular design adapts to various crops like chili, tea, or cotton.

Agronomic Value

Digital Literacy

IoT dashboards enhance farmers' digital skills.

Health & Safety

Reduced dust and fatigue improve farmer well-being.



Farmer Empowerment

IoT tools are easy to use and accessible for all farmers.

Rural Employment

Local manufacturing and maintenance create jobs in rural areas.

Social Impact

Who it Impacts?

Small & Marginal Farmers

- Gain access to affordable mechanization (< ₹20,000).
- Save 60–70% labor effort and increase income through higher-quality lint.

Women & Elderly Workers

- Portable, ergonomic design enables safe, low-fatigue operation.
- Expands participation in agricultural labor and rural productivity.

Rural Entrepreneurs & MSMEs

- Local manufacturing, assembly, and service opportunities.
- Drives Atmanirbhar Bharat by enabling decentralized rural production.

Environment & Community

- Solar-powered operation reduces emissions and promotes green farming.
- Anti-static and low-heat materials ensure lint safety and prevent fire hazards.

Government & Policy Makers

- Directly supports Atmanirbhar Bharat, Digital India, and Sustainable Agriculture Missions.
- Provides data-driven insights for crop policy planning and yield forecasting.

Climate Resilience

Minimizes emissions and enhances climate adaptability.

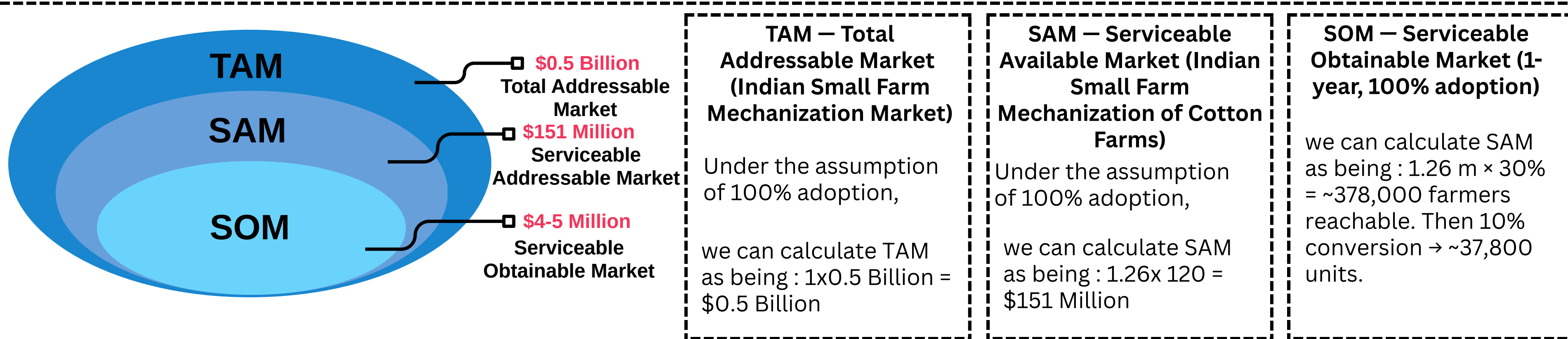
Solar-Battery Hybrid

Reduces fossil fuel use and lowers operational expenses.

Technological Trailblazer

Packaging Marvel

Immediate packaging of the cotton that's been picked



REFERENCE

- 1) J. M. Bennett, N. P. Woodhouse, T. Keller, T. A. Jensen and D. L. Antille, "Advances in cotton harvesting technology: a review and implications for the John Deere round baler cotton picker," Journal of Cotton Science, vol. 19, no. 2, pp. 225-249, 2015.
- 2) J. Bankar, G. Nimbalkar, M. Nimbalkar, A. Hon, S. Ghodke and S. Tamboli, "Design & Development of Smart Cotton-Picking Robot Using ML & Suction Mechanism," 2025 SAE Media Group TechBriefs Contest – Robotics & Automation, June 1 2025.
- 3) C. Zhang et al., "The Development of Wear Characteristics of the Picking Hook Teeth," Wear, vol. XXXXX, 2024. doi:10.1016/j.wear.2024.00060.

- 4) A. D. Abdazimov, S. S. Radjabov and N. N. Omonov, "Automation of agrotechnical assessment of cotton harvesting machines," Journal of Physics: Conference Series, vol. 1260, no. 3, p. 032001, 2019. doi:10.1088/1742-6596/1260/3/032001.

LINKS

[1\) Documentation](#)

[2\) TAM SAM SOM Documentation](#)

[3\) Blender Model Images](#)

[4\) Budget Report](#)

[5\) Demo Video Link - Web Demo](#)

[6\) Actual Website](#)